



INFRASTRUCTURE

Disaster Recovery Manual

Recovery Procedures



- Disaster Recovery High-Level Diagram
- Disaster Recovery Organization Chart
- In-Scope Servers
- Network Mappings
- Actual recovery procedures for:
IBM i-Series, Databases, ESSBase, TMS,
Lennar.com, Informatica, Metaviewer, Telecom
- Lennar VoIP Topology
- VMware Site Recovery Manager

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Release Date: 05-15-20

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VERSION HISTORY

Version	Date	Name	Comments/Changes
1.2	05-07-18	Shilpa Kaul	Created initial draft of the document.
1.2	10-25-18	Ricardo Sabates	Reviewed the document.
1.2	10-26-18	Shilpa Kaul	Document released.
2.0	05-14-20	Ricardo Sabates	Reviewed the document.
2.0	05-15-20	Shilpa Kaul	Document released.

DOCUMENT APPROVER

Document Approver	Designation	Signature	Approval Date
xxx	Infrastructure Architect		

CONTRIBUTORS

Contributor	Designation
xxx	Infrastructure Architect
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ABBREVIATIONS & ACRONYMS

Term	Description
ACL	Access Control List
CRG	Cluster Resource Group
DNS	Domain Name System
DSS	Distributed Systems and Storage
FTP	File Transfer Protocol
FSMO	Flexible Single Master Operations
LTS	Lennar Technology Solutions
LUN	Logical Unit Number
MDC	Miami Data Center
NIC	Network Interface Card
RDA	Remote Diagnostic Agent
RPA	Robotic Process Automation
RTO	Recovery Time Objective
SSH	Secure Shell
TMS	Treasury Management System
VM	Virtual Machine

OVERVIEW

Disaster Recovery (DR) is an area of security planning that aims to protect an organization from the effects of significant negative events. Disasters could include loss of power, system failure, fire, hurricanes, and so on.

This document aims to list Lennar's IT infrastructure and the steps that Lennar will follow if a disaster occurs. This will help reduce both the threat and impact, should the worst transpire. To accomplish this, the document captures relevant information about Lennar's ability to recover its IT systems after a disaster situation.

PURPOSE

The purpose of this document is to assess Lennar's ability to restore its core systems from Lennar's primary datacenter located in Miami, FL to the alternate datacenter in Las Vegas, NV.

IN SCOPE APPLICATIONS

Following is the list of applications in-scope:

- JDE
- Showcase
- ESSBASE
- Hyperion
- Treasury Management System (TMS)
- Metaviewer
- Lennar.com (Website)
- Informatica

AUDIENCE

The intended audience of this document are the personnel in Lennar Technology Solution (LTS) in the following departments only:

- DSS
- Database
- IBM i-Series
- Telecom
- Networking
- Application/Product Support Teams

RECOVERY SCENARIOS

Planned migration is the orderly migration of virtual machines from the protected site (MDC) to the recovery site (LAS). Planned migration prevents data loss when migrating workloads in an orderly fashion. For planned migration to succeed, both sites must be running and fully functional.

DISASTER RECOVERY HIGH-LEVEL DIAGRAM

The disaster recovery system consists of two sites: a primary server and a recovery server. During a disaster, all operations are diverted to the recovery server so that data and service availability are continued.

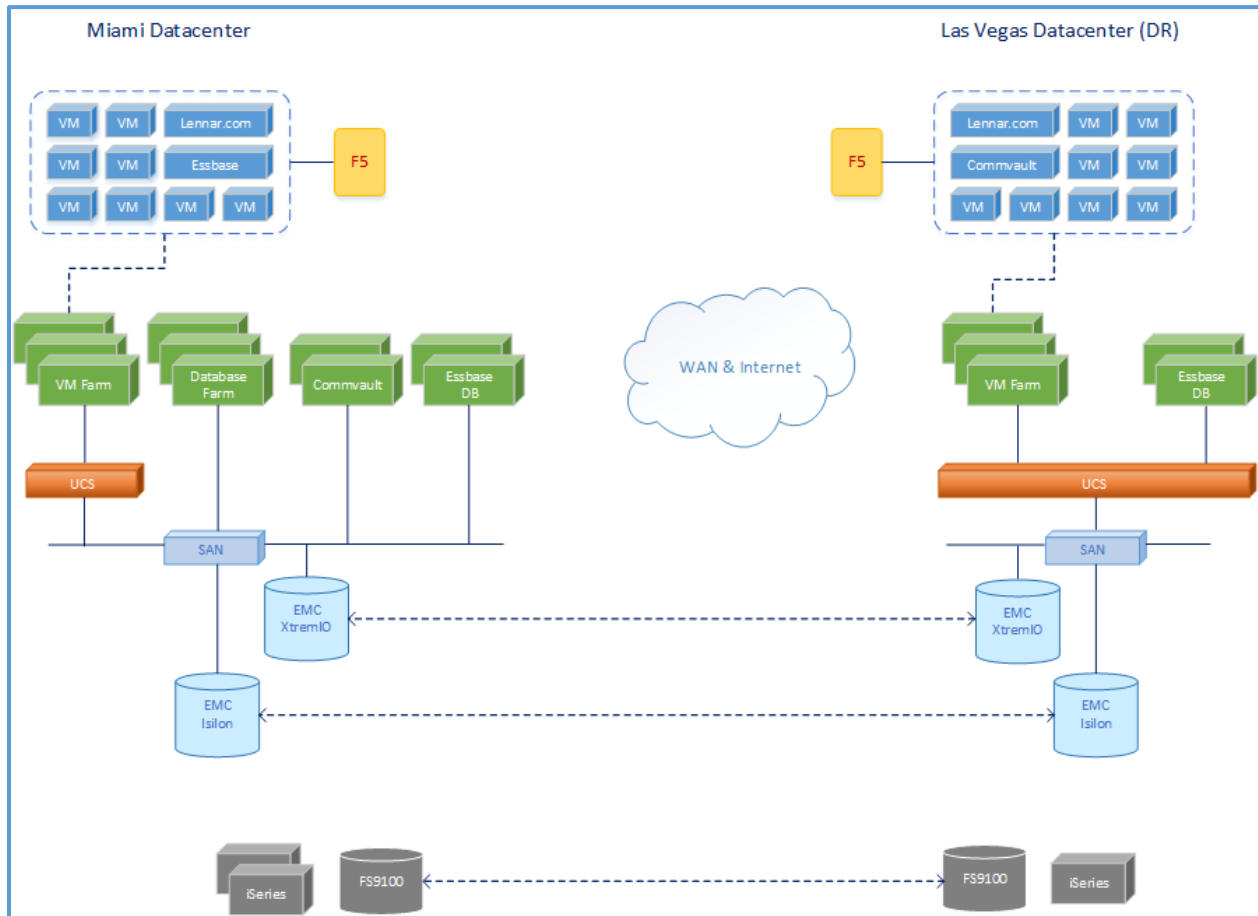


Figure 1: Disaster Recovery High-Level Diagram

DISASTER RECOVERY INFRASTRUCTURE CAPABILITIES**Recovery Time Objective (RTO)**

Recovery Time Objective (RTO) is the expected length of time and a service level within which your business must be reestablished after a disaster in order to avoid undesirable consequences related with a disruption in business continuity.

System	Supports	Recovery Time	Replication Method
JDE	Lennar Corp/Rialto/NewHall/ SunStreet/FivePoint/UAMC/NAT	8 hours	Continuous Replication
TMS	Lennar Corp	8 hours	Continuous Replication
SHOWCASE	Lennar Corp/ Rialto	8 hours	Daily Replication
ESSBASE/Hyperion	Lennar Corp	8 hours	Continuous Replication
CUCM(Phones)	Lennar/Rialto/UAMC/NAT	8 hours	Continuous Replication
Telepresence (Video)	Lennar/Rialto	48 hours	Daily Backup
Lennar.com	Lennar Corp	8 hours	Continuous Replication

CONTACT DETAILS

DSS

Point of Contact	Satellite Phone?	Phone Number

Open to Travel

Database

Point of Contact	Satellite Phone?	Phone Number

Open to Travel

Name	Phone Number

IBM i-Series

Point of Contact	Satellite Phone?	Phone Number

Open to Travel

Name	Phone Number

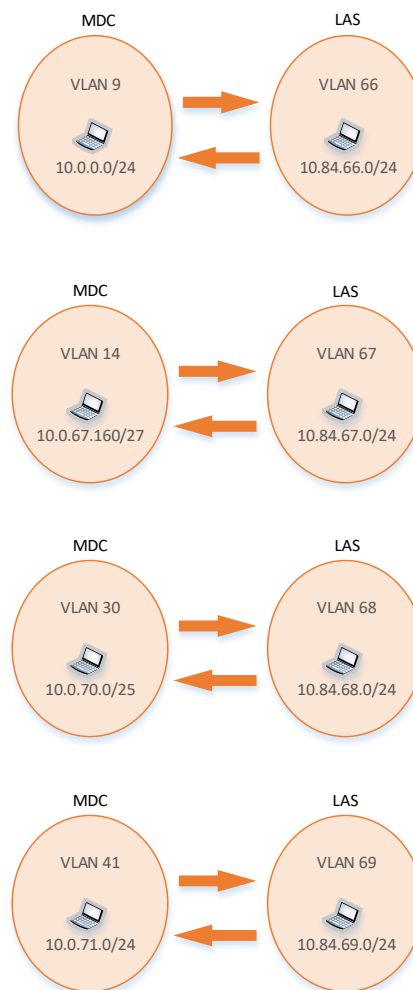
IN-SCOPE

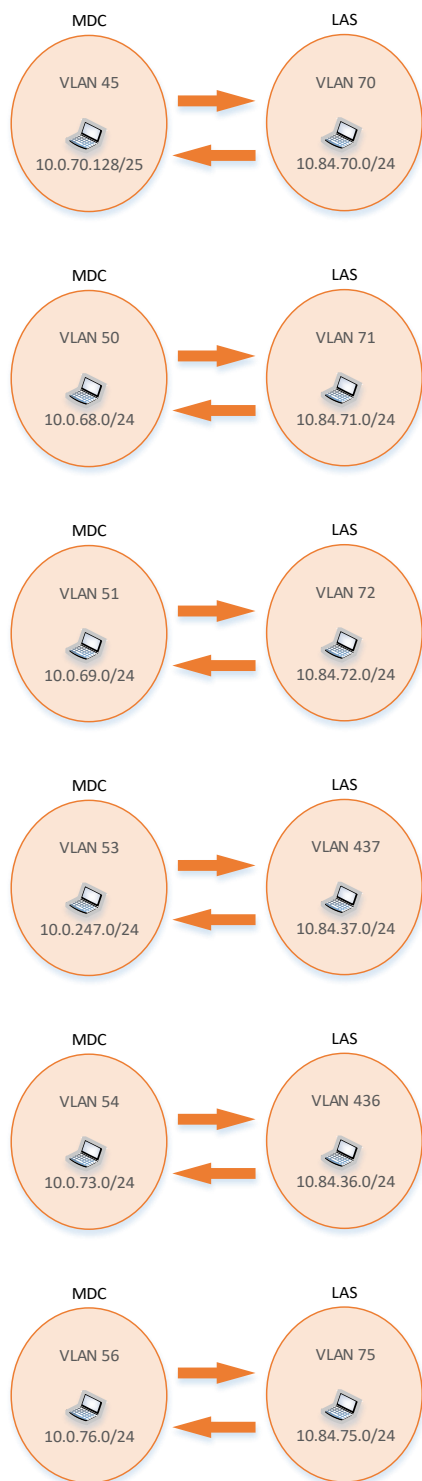
SERVICES

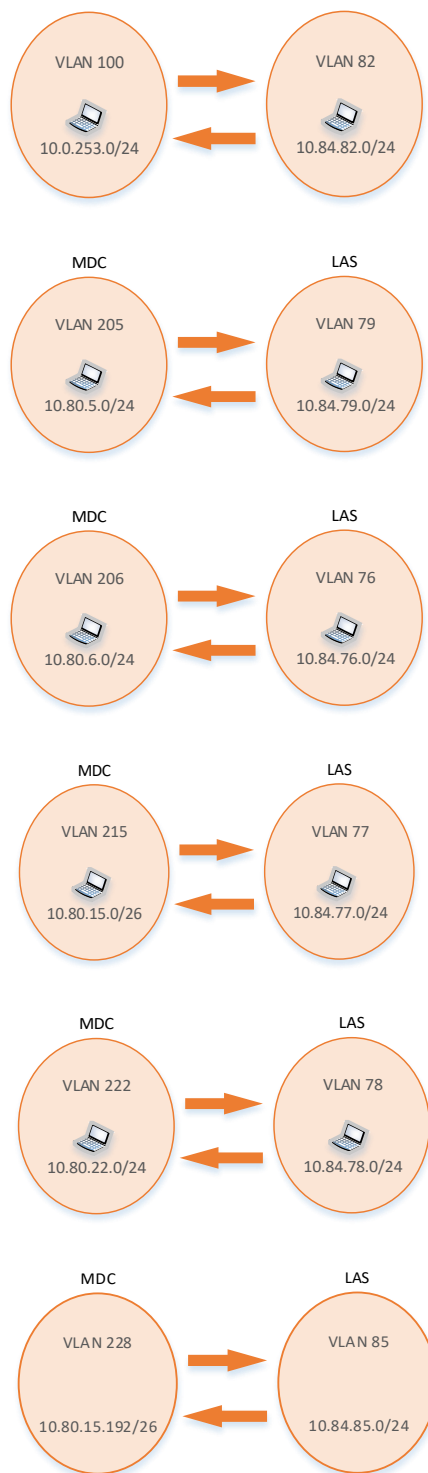
NETWORK MAPPINGS

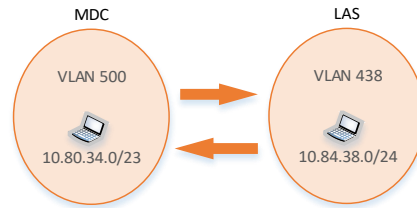
Network mapping supports to control network configuration for the VMs. Network mapping can be specified between the source and the destination network. In case of failover, the VMs are registered with the network configuration that is specified in the network mapping.

Below is the pictorial representation of the network mappings between MDC and LAS:









RECOVERY FLOW – ACTUAL RECOVERY

The actual recovery flow is listed as follows:

1. Determine the applications targeted for recovery.
2. Initiate IBM i-Series recovery.
3. Recover shared databases by:
 - a. Site Recovery Manager.
 - b. Forceful recovery of surviving node.
4. Perform failover via Site Recovery Manager for each application VM recovery plan.
5. Recovery ESSBASE physical server via RPA/UCS.
6. DNS updates, if applicable.
7. Telecom.

ACTUAL RECOVERY PROCEDURES

IBM I-SERIES

Switch Instructions – Update DNS

Record	IP address	A Record	CNAME



Only the 'A Records' need to be updated during a failover event.

Switch Instructions – Switch Preparation**RETURN OPERATIONS TO MIAMI**

From infrastructure perspective, to go back to Miami Data center for all DSS, Database, Lennar.com, replication should be stopped.

FREQUENTLY ASKED QUESTIONS

1. SRM was unable to change an IP address of a VM during recovery. What should I do?

Answer - Login in using the local admin account via VM console and update manually. Cylance, an out-of-date VM agent, or an unsupported OS may cause this.

APPENDIX A: VMWARE SITE RECOVERY MANAGER

VMware Site Recovery Manager is a disaster recovery solution that helps plan, test, and run the recovery of virtual machines between Lennar's primary data center (MDC) and the recovery data center (LAS). Site Recovery Manager can be used to implement different types of recovery from MDC to LAS.

Disaster recovery does not require both data centers to be up and running, and it can be initiated in the event when MDC goes offline unexpectedly.

- Site Recovery Manager harmonizes the recovery process with the replication mechanisms, to minimize data loss and system downtime
- Site Recovery Manager lets you test recovery plans. Tests can be conducted by using a temporary copy of the replicated data in a way that does not disrupt ongoing operations at either site